

Summary of hyp VQVAE

Preliminary Setup

- The manifold is modeled as a hyperbolic space $\mathbb{H}^n$. A point is described by its radius-angle representation: radius $r = \text{dist}(x, 0)$, angle difference $\Delta\theta$.

- The quantization objective is to minimize the nearest neighbor distance (or its square):

$$\mathbb{E}[\min_i d(x, c_i)^p] \quad (p = 1 \text{ or } 2),$$

together with a commitment term.

- The number of codewords (K directions) is limited; training must simultaneously optimize both encoder output and codeword locations.

Key Formula (Distance = Geodesic Guidance)

For two points x, y , the hyperbolic distance (in polar coordinates) is:

$$\cosh d(x, y) = \cosh r_x \cosh r_y - \sinh r_x \sinh r_y \cos(\Delta\theta).$$

If $r_x = r_y = R$, then:

$$\cosh d = 1 + 2 \sinh^2 R \cdot \sin^2(\Delta\theta/2) \Rightarrow \\ d(R, \Delta\theta) \approx 2R + 2 \log \sin(\Delta\theta/2) \quad (R \text{ large}).$$

Takeaways:

- With fixed angle difference $\Delta\theta$, distance grows approximately linearly with radius R (derivative $\partial d / \partial R \approx 2 > 0$).
- If the distance is purely radial, $d^2 \approx 4R^2$ (radial growth), so larger R intensifies the "radial expansion" effect.

Proposition 1: Radial Expansion Prioritizes Angular Coverage (Structural Inner Bias)

During training with a limited number of codewords, to maximize coverage of angular diversity, all base radii can be scaled outward to a larger R . This increases pairwise distances between codewords of different angles ($\approx 2\delta$), while not

harming angular separation. In contrast, optimizing angular coverage (increasing angular resolution) can only achieve logarithmic improvement.

Conclusion:

With limited codewords, moving all codewords radially outward to the same radius is a globally consistent improvement strategy.

- This produces a consistent “expansion bias” during training: codewords keep drifting outward to larger R .
- If the encoder output and commitment term are jointly optimized, both will drift outward, forming a “radial expansion” phenomenon.

Proposition 2: Fréchet Mean Bias in NPC Space

Hyperbolic space is an NPC (non-positive curvature) space, where distance squared is convex (Busemann convexity). For points distributed at the same radius $r = R$ with angular spread, the Fréchet mean will lie near the origin (extreme minimizer).

Implications:

- In all types of “K-means / K-medoids” updates (or EMA codebook updates), the mean will always be pulled inward.
- Thus, clustering/quantization centers will always drift toward the origin, while codewords drift outward → producing an inherent “tug-of-war.”

Proposition 3: Angular Resolution Scaling with Radius

The volume of a spherical shell at radius R is:

$$\text{vol}(B(0, R)) \sim \int_0^R \sinh^{n-1}(t) dt \approx \frac{1}{n-1} e^{(n-1)R}.$$

Consequences:

- Covering the angular region on a large-radius sphere requires exponentially many codewords. For limited codewords, this is impossible.
- To reduce quantization distortion, the system tends to shrink the base radius R , because at smaller R , the same number of codewords can cover a proportionally larger angular region (see the $\log \sin$ term earlier).
- Therefore, encoder outputs closer to the origin yield smaller distortion.

Proposition 4: Geodesic Shrinkage Induces Distortion (Color/Contrast Collapse)

When radius grows, hyperbolic geodesic distance approximates linear in radius. For the same angular differences, distances blow up:

- Angular diversity gradually loses effective contribution ("color/contrast collapse" in visualization).
- Codewords collapse into fewer distinct classes; output embeddings lose angular variety.
- Training may further reinforce this collapse: frequent updates + radial shrinkage push outputs toward "mean collapse."

So in short:

- **Radial drift outward:** maximizes angular coverage (bias).
- **Fréchet mean inward:** pulls encoder outputs inward (bias).
- **Codebook limitation + exponential shell volume:** encourages shrinking radius.
- **Training dynamics:** causes gradual collapse of angular diversity → loss of contrast/semantic variety.

Even if I add regularization/clever tricks, it's still hard to cure the root cause

I've been trying to summarize the observations (why variance collapse and mode collapse in hyperbolic VQ are hard to fix):

- **Increasing sampling diversity (e.g., using more frequent negatives, multiple loss terms):** can only make training *look* more diverse, but it does not change the fact that "radial shrinkage is the most efficient way to globally reduce distance."
- **Radial shrinkage as an auxiliary objective** (fixing target radius, enlarging or reducing angular deviation): it acts as a side penalty (not the real target), but its gradient can easily dominate the training dynamics; the more you emphasize it, the more angular resolution you sacrifice, and finally the system degenerates to collapse.

- **The more important regime is actually near the boundary:** even short geodesics near the boundary show the same monotonic inward shrinkage; adding penalties only shifts the rate.
- **System-level stability:** The system “chooses” to shrink radii collectively because this speeds up and stabilizes convergence; but there is no constraint to preserve angular resolution, so the collapse is a structural bias.
- **Radial bins + angular codebook** resembles clustering: essentially falls into Euclidean-style clustering with radial weighting, but in hyperbolic space the same issues remain—angular discrimination collapses, and the polar form only looks like clustering, without improving objective separation.

Mini-summary (Why “vanilla” VQ is not suitable for hyperbolic space)

- In hyperbolic $\mathbb{H}^n$, distances grow almost linearly with radius (or squared distance), which makes the system tend to reduce distances globally by shrinking all radii—a general failure mode of quantization.
- The **NPC property** (non-positively curved space) ensures convexity along geodesics, so all updates (EMA, centroid moves, etc.) are biased inward, further encouraging radial collapse.
- Geodesic distances prefer radial shrinkage: when radius is large, angular separation contributes very little compared to radial terms, thus the system naturally ignores angular resolution.
- These factors cause: even if you carefully initialize, with training the basis vectors (codes) shrink radially, angular separation vanishes, and reconstructions lose fine detail, leading to **structural collapse**.
- Therefore: hyperbolic space is not “suited for vanilla VQ.” You need to redesign the target/objective so that the codebook no longer aligns with radial shrinkage. Otherwise, even with clever tricks, collapse remains inevitable.

What conditions are necessary for success?

If you really want VQ in hyperbolic space to work, you must add at least one of the following interventions (not just parameter tuning):

1. **Radial-aware weighting of the objective**, e.g. scaling distance penalty as $e^{-\alpha r(x)} d(x, y)$, so that codes at larger radius are less penalized for angular differences.
2. **Geodesic-volume-based normalization**: weight by $\sinh^{n-1} R$, so that codes at larger radius are correctly accounted for by density.
3. **Radial binning + angular separation**: explicitly constrain codes so that radii are preserved while only angular separation is optimized (as in spherical codebooks).

If none of these are applied, then the system will always prefer to shrink the entire codebook inward. As a result, your hyperbolic VQ-VAE (or variants) will *inevitably* collapse: codes cluster near the origin, angular resolution vanishes, and the quantizer degenerates.

Possible next step

1. Distance Reweighting in Main Loss (Primary Fix)

Problem: The hyperbolic distance's linear growth with radius makes "uniform inward contraction" the globally optimal strategy for minimizing quantization error.

Solution: Introduce radius-dependent reweighting in the main quantization loss to counteract this bias.

Key Insight: By downweighting distances for large-radius pairs, we make "staying at large radius" relatively more attractive than "contracting inward."

2. Measure-Based Codebook Distribution Matching (Structural Constraint)

Problem: The hyperbolic volume grows exponentially with radius, but finite codebook size K cannot maintain sufficient angular resolution at large radii.

Solution: Enforce that the empirical codebook usage distribution matches the theoretical hyperbolic measure distribution.

Key Insight: This directly penalizes deviations from the hyperbolic measure, making "radius collapse" increase the main loss rather than decrease it.

3. Tangent Space Quantization with Radius Constraints (Architectural Fix)

Problem: The unified nearest-neighbor search in hyperbolic space inherently favors radius contraction.

Solution: Decouple angular and radial quantization by working in tangent space for angular components while explicitly controlling radius.

Key Insight: By quantizing in tangent space (Euclidean) for angular components and explicitly controlling radius, we avoid the inherent "radius contraction" bias of hyperbolic nearest-neighbor search.

4. Measure-Aware Initialization and Reset (Prevention)

Problem: Standard uniform initialization and dead code reset don't account for hyperbolic measure, leading to poor initial distribution.

Solution: Initialize and reset codebook vectors according to the hyperbolic measure distribution.

5. Radius-Gated Temperature Scheduling (Adaptive Control)

Problem: Fixed temperature in soft quantization doesn't account for the varying difficulty of angular resolution at different radii.

Solution: Use radius-dependent temperature to encourage angular diversity at large radii while allowing concentration at small radii.